# POLISCI 158 – AI & Politics

## Course Information

Fall Quarter 2026
Variable Units: 5 units (Full Credit), 4 or 3 units (Reduced Credit)
Monday – Wednesday, 3:00 – 4:20 pm
Location: Thornton Center, Rm 110
Weekly Sections: begin in Week 2 (week of Sep 28); Encina West 202 (times listed under Course Teaching Assistants)

## Course Instructors

Simone Paci, Ph.D.
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Meeting hours: Sign-up here

Yiqing Xu, Ph.D.
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## Course Teaching Assistants

Xinru Pan
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Meeting Hours: TBD
Section 1: Monday, 10:30 – 11:20 am
Section 2: Wednesday, 10:30 – 11:20 am
Location: Encina West 202

Zihan Xie
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Meeting Hours: TBD
Section 3: Friday, 9:30 – 10:20 am
Section 4: Friday, 11:30 am – 12:20 pm
Location: Encina West 202

## Course Overview

Artificial intelligence is increasingly both a political object and a tool for political and social life. This course introduces students to the core concepts behind AI systems while examining their political, social, and institutional consequences. The course combines the politics of AI — how AI reshapes power, resources, governance, markets, elections, labor, infrastructure, and global competition — with AI for politics — how AI tools can be responsibly used in research, policy analysis, civic work, public administration, advocacy, journalism, and other social applications.

Students will learn the basic intuitions behind machine learning, generative AI, large language models, training data, evaluation, bias, and uncertainty, without requiring advanced programming or computer science preparation. They will also analyze real-world applications and conflicts involving AI in public opinion, campaigns, bureaucracies, regulation, firms, data centers, geopolitics, and social-service delivery. By the end of the quarter, students will be able to explain core AI technologies, assess their political and social implications, evaluate opportunities and risks in applied settings, and design a project that uses or studies AI in relation to a concrete social or political challenge.

The course is co-taught and runs two parallel tracks. Mondays (Prof. Paci) examine the political economy of AI: who demands it and to what political effect, who produces it and on what terms, and the infrastructural and geopolitical conflicts that follow. Wednesdays (Prof. Xu) build applied understanding of how AI systems

work and how to use them well — deep learning, natural language processing, computer vision, generative AI and large language models, post-training, and agents. The two tracks are sequenced independently and meet in the final project. Three sessions are taught jointly: the introduction and the two final presentation sessions.

## Course Timeline

| Week | Date | Topic |
|---|---|---|
| Week 1 | Wed, Sep 23 | Introduction: the politics of revolutionary technological change; GenAI 101 |
| Week 2 | Mon, Sep 28 | Politics — Demand side I: labor markets, productivity, automation |
| | Wed, Sep 30 | Tech — Statistical Learning and Deep Learning |
| Week 3 | Mon, Oct 5 | Politics — Demand side II: firms, platforms, market structure, lobbying |
| | Wed, Oct 7 | Tech — Natural Language Processing |
| Week 4 | Mon, Oct 12 | Politics — Demand side III: public opinion, beliefs, and the information environment |
| | Wed, Oct 14 | Tech — Large Language Models |
| Week 5 | Mon, Oct 19 | Politics — Supply side I: politicians, campaigns, and elections |
| | Wed, Oct 21 | Tech — Post-Training and Reasoning Models |
| Week 6 | Mon, Oct 26 | Politics — Supply side II: public institutions, service delivery, and governance capacity |
| | Wed, Oct 28 | Tech — Computer Vision and Multimodality |
| Week 7 | Mon, Nov 2 | Politics — Data Centers: siting, energy and water, local politics |
| | Wed, Nov 4 | Tech — Agentic AI: From Chatbots to Agents |
| Week 8 | Mon, Nov 9 | Politics — US–China Competition: compute, chips, export controls |
| | Wed, Nov 11 | Tech — AI for Politics: Democratic and Authoritarian Uses |
| Week 9 | Mon, Nov 16 | Politics — Data Sovereignty: diffusion, data governance, state capacity |
| | Wed, Nov 18 | Tech — AI Agents: Reliability, Safety, and Governance |
| | Nov 23 & 25 | Thanksgiving recess — no class |
| Week 10 | Mon, Nov 30 | Final Presentations — themed panels |
| | Wed, Dec 2 | Final Presentations — themed panels; course wrap-up |

## Course Learning Goals

1) Explain the core concepts behind modern AI systems — machine learning, deep learning, natural language processing, computer vision, large language models, post-training, and agents — at an applied, intuitive level.

2) Use AI tools effectively and responsibly — prompting, evaluation, APIs, and human–AI workflows — and recognize their failure modes, including hallucination, bias, and overreliance.

3) Analyze how AI redistributes power, resources, and legitimacy, using core political economy frameworks and distinguishing what is genuinely new about generative AI relative to earlier digital and automation technologies.

4) Assess emerging empirical trends in the politics of AI, and the quality and limits of the available evidence.

5) Design and execute a project that uses or studies AI in relation to a concrete social or political challenge.

6) Present and critique technical and political work for a mixed audience — presenting findings and giving

usable peer feedback.

## Course Requirements

POLISCI 150A (Introduction to Data Science), or equivalent exposure to programming in Python or R, is recommended but not required. Curiosity about politics and technology is essential. The applied portions of the course involve light Python — mainly calling APIs — and support is provided for students without prior programming experience; no advanced programming is expected. The course remains at an applied level, and students seeking deep technical training in algorithms should consider computer science courses.

Students are required to have access to a frontier large language model product: OpenAI ChatGPT Plus, Google Gemini Pro (search "Gemini Pro for students" for a free one-year trial), Anthropic Claude Pro, or an equivalent.

We may occasionally ask students to use paid AI products and services for particular tasks. The teaching staff will be mindful of the potential costs and will make every effort to identify free or low-cost alternatives, including the Stanford AI Playground, whenever possible. Contact us if cost is a barrier.

## Student Expectations

### Weekly Preparation (before class)

Each week there will be material to review in preparation for class. This may include academic papers, news articles, reports, podcasts, or blog posts. All materials marked with an asterisk * on Canvas are required; the rest are suggested. Come with questions from the assigned material.

### Class Participation

Class time combines lecture and demonstration with seminar-style discussion, worksheets, and short exercises. Students are expected to engage actively, bring questions from the assigned material, and provide constructive feedback on peers' work. Cold calling is to be expected. Students are also expected to hold at least one individual meeting with the professors or the teaching assistants during the quarter.

### Weekly Section

One TA-led section meets each week, starting in Week 2 (week of September 28), and carries the hands-on work of the course: API calls, small data exercises, tool building, and structured critique. Each section includes a graded activity or assignment. The grading system for these assignment will be Pass/No-Pass.

### Peer Review

After each project submission, students review one classmate's milestone and provide written feedback. Peer feedback is also given live at the final presentation panels. Giving usable feedback is a course learning goal, and it is graded.

## Project-Based Assessments

Students will select a concrete social or political challenge and build a project around it, either using AI on that challenge or studying AI's effect on it. Projects may be theoretical, empirical, argumentative, or applied, and they may draw on either or both course tracks.

The project has three graded submissions:

- Proposal (question, motivation, feasibility)
- Implementation milestone (design, evidence, or build progress)
- Final submission (integrated writeup plus attachments)

The final presentation panels take place in person on Monday, November 30 and Wednesday, December 2.

## Grading

| 5 Units | 4 units | 3 units | Weight |
|---|---|---|---|
| **Presentation** | | | **15%** |
| Final presentation | Final presentation | Final presentation | 15% |
| **Final Project** | | | **50%** |
| Proposal milestone | 20% shorter | 30% shorter | 10% |
| Implementation milestone | 20% shorter | 30% shorter | 10% |
| Final submission | 20% shorter | 30% shorter | 30% |
| **Section Activities** | | | **25%** |
| Section Assignment (8) | Section Assignment (7) | Section Assignment (6) | |
| **Peer Review** | | | **10%** |
| Milestone reviews (3) | Milestone reviews (2) | Milestone review (1) | |
| **Total** | | | **100%** |

**Letter Grade Conversion Scale**

| F | D– | D | D+ | C– | C | C+ | B– | B | B+ | A– | A | A+ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| <60 | 60–62 | 63–66 | 67–69 | 70–72 | 73–76 | 77–79 | 80–82 | 83–86 | 87–89 | 90–92 | 93–96 | 97+ |

## Final Project Guidelines

Each student will complete a focused final project on AI and politics. The project may use AI as a tool on a political or social problem, or study AI's political, institutional, or distributive effects. Projects should engage course concepts from at least one of the two tracks, and the strongest projects usually engage both.

## Project Components and Deadlines

| Component | Due Date | Expectations | Min. Requirements |
|---|---|---|---|
| Proposal Milestone | Sun, Oct 11 (11:59 p.m.) | Motivate the project; explain feasibility | 300–500 words (min–max) |
| Peer Review 1 | Sun, Oct 18 (11:59 p.m.) | Written feedback on one classmate's proposal | 150–200 words |
| Implementation Milestone | Sun, Nov 15 (11:59 p.m.) | Design, evidence, or build progress; what has actually been done | 800–1000 words |

| Component | Due Date | Expectations | Min. Requirements |
|---|---|---|---|
| Peer Review 2 | Sun, Nov 22 (11:59 p.m.) | Written feedback on one classmate's milestone | 150–200 words |
| Final Presentation | Mon, Nov 30 / Wed, Dec 2 | Themed panel presentation and Q&A | 5 minutes; 3 slides |
| Peer Review 3 | In session, Nov 30 / Dec 2 | Structured feedback on a panel peer | 150–200 words |
| Final Submission | Finals week | All components integrated, plus attachments | 2000–3000 words |

## Course Policies

### Absences

The course is highly interactive and participation is key to its pedagogy and learning objectives. However, life happens and sporadic absences can be excused. Students can make up a missed session through a short reflection on the week's material or a brief peer-review memo.

### Extensions

The course follows a steady pace and is structured to even out student work. We strongly encourage students not to delay assignments. That said, we will consider personal circumstances in granting extensions when requested in advance.

### Late Submission

Work must be submitted electronically to Canvas by the stated deadline. Except for extraordinary circumstances, late submissions will be penalized by a deduction of 0.5 points from the final course grade for each day late. Deadlines for the final presentations and peer reviews are firm, and late submissions will not be accepted.

### Re-Grading

Each graded assessment has clearly stated expectations and grading rubrics. The margin for re-grading will be slim. Students who believe their work deserves a major letter-grade change can bring this up through office hours or email.

### Instructor Communication

Reach out for questions, doubts, or problems. We welcome feedback and strive to incorporate student thoughts into our teaching practice. Email works best. We respond promptly during weekdays but may not always be available on weekends.

### Creating a respectful environment

This course encourages lively and earnest discussion on political topics. Disagreement is inevitable; mutual respect is non-negotiable. Please listen attentively, critique ideas rather than persons, and help create space for others to contribute.

### Inclusivity

This course strives to include a diverse array of authors, opinions, cultural and intellectual backgrounds. Similarly, the assessment strategy — split across presentations, written work, activities, and peer review —

recognizes that students may hold different strengths due to individual and/or cultural differences, and gives them several routes to demonstrate what they have learned.

**Using Technology in the Classroom**

During lecture, presentation, and discussion segments, the use of laptops and electronics is not allowed. During worksheets, coding, and other activity segments, devices are expected: you will need them to consult material, run code, and complete the activity. Each session will announce its segments.

**Generative AI**

Because this course both studies and uses AI, we hold the line explicitly. You are encouraged to use AI tools extensively for the applied work: exploring, prototyping, debugging, building, and analyzing. All written assignments, however, must be in your own words and must reflect your own understanding. You must disclose substantive AI use in a brief AI-use note attached to the submission, crediting tools the way you would credit a collaborator. Using AI writing tools and submitting the output as your own violates the Stanford Honor Code.

**Academic Integrity**

Academic integrity is vital to a supportive and engaging learning environment. Please review Stanford's Honor Code and documentation and citation resources from the Hume Center for Writing and Speaking. When in doubt, cite and ask.

## Academic Accommodations

Students who may need an academic accommodation based on the impact of a disability must initiate the request with the Student Disability Resource Center (SDRC) within the Office of Accessible Education (OAE). SDRC staff will evaluate the request, recommend reasonable accommodations, and prepare an Accommodation Letter for faculty. Students should contact SDRC as soon as possible since timely notice is needed to coordinate accommodations.

OAE location: 563 Salvatierra Walk (650-723-1066). For more information: http://studentaffairs.stanford.edu/oae

## Title IX

Title IX is a federal law that prohibits discrimination on the basis of sex and gender, including sexual harassment and sexual violence, in educational programs and activities that receive federal financial assistance. Stanford is committed to providing a campus environment free of such discrimination. Many employees of Stanford, including course instructors, are required to report Title IX-related incidents.

**Confidential Resource:**

The Confidential Support Team (CST) is reachable via a 24-7 hotline at 650-725-9955 for urgent support, or at 650-736-6933 for non-urgent questions and appointments.

**Non-Confidential Resource:**

The Title IX Office is reachable at 650-397-4955 or via email at titleix@stanford.edu. The Office of Sexual Assault and Relationship Abuse Education and Response (SARA) is reachable at 650-725-1056 and

saraoffice@stanford.edu.

More information is available online at: sexualviolencesupport.stanford.edu, relationshipviolencesupport.stanford.edu, transgendersupport.stanford.edu

## Academic Disruptions

Stanford is committed to the highest quality education. As your instructors, our priority is to uphold your educational experience, including through short- or long-term disruptions (public health emergencies, natural disasters, protests, demonstrations). If adjustments are necessary, we will communicate clearly and promptly to ensure expectations remain clear and achievable.

## Preliminary Course Schedule and Recommended Readings

### Session 1 – Wed, Sep 23 – Introduction (Paci & Xu)

**Class Topics**

- Course overview, expectations, and how the two tracks fit together
- Politics 101: power, institutions, winners and losers, state capacity, legitimacy, regulation
- What makes a technology politically transformative?
- GenAI 101: models, capabilities, limits, and why diffusion may concentrate

**Recommended Readings**

* To be posted on Canvas

### Session 2 – Mon, Sep 28 – Politics: Demand Side I — Labor Markets, Productivity, Automation (Paci)

**Class Topics**

- Task substitution vs. augmentation; the distributional politics of task change
- The political economy of displacement and adjustment

**Recommended Readings**

* To be posted on Canvas

### Session 3 – Wed, Sep 30 – Tech: Statistical Learning and Deep Learning (Xu)

**Class Topics**

- What it means for a machine to "learn": rules from examples, training and test data, overfitting
- Neural networks and deep learning, conceptually; where training data comes from

**Recommended Readings**

* James, Witten, Hastie, Tibshirani, and Taylor (2023), *An Introduction to Statistical Learning (Python edition)*, section 2.1. Free at statlearning.com.
* 3Blue1Brown (2017), "But what is a neural network?" (19 minutes). https://www.youtube.com/watch?v=aircAruvnKk

### Session 4 – Mon, Oct 5 – Politics: Demand Side II — Firms, Platforms, and Lobbying (Paci)

**Class Topics**

- Market structure: platforms and the concentration of capabilities
- Lobbying, standard setting, and policy influence
- Open vs. closed models and governance coalitions

**Recommended Readings**

* To be posted on Canvas

**Session 5 – Wed, Oct 7 – Tech: Natural Language Processing (Xu)**

**Class Topics**

- Tokens, embeddings, and attention: how text becomes numbers
- Text as data in political science, and what censored text teaches a model

**Recommended Readings**

* Financial Times (2023), "Generative AI exists because of the transformer" (visual essay). https://ig.ft.com/generative-ai/
* Grimmer and Stewart (2013), "Text as Data," *Political Analysis* 21(3), the opening section and its four principles.

**Session 6 – Mon, Oct 12 – Politics: Demand Side III — Public Opinion and the Information Environment (Paci)**

**Class Topics**

- Beliefs, trust, and the information environment
- Civil society, creators, and knowledge institutions
- AI and democratic deliberation

**Recommended Readings**

* To be posted on Canvas

**Session 7 – Wed, Oct 14 – Tech: Large Language Models (Xu)**

**Class Topics**

- Predicting the next word at scale: pretraining, scaling, open and closed models
- Context and prompting; why models make things up

**Recommended Readings**

* 3Blue1Brown (2024), "Large Language Models explained briefly" (8 minutes). https://www.youtube.com/watch?v=LPZh9BOjkQs
* OpenAI (2025), "Why language models hallucinate" (blog post on Kalai et al.). https://openai.com/index/why-language-models-hallucinate/

**Session 8 – Mon, Oct 19 – Politics: Supply Side I — Politicians, Campaigns, and Elections (Paci)**

**Class Topics**

- GenAI in political communication: persuasion, microtargeting, fundraising
- Synthetic political media and disclosure regimes
- Election administration and integrity challenges

**Recommended Readings**

* To be posted on Canvas

**Session 9 – Wed, Oct 21 – Tech: Post-Training and Reasoning Models (Xu)**

**Class Topics**

- Fine-tuning and reinforcement learning from human feedback: how a text predictor becomes an assistant
- Written rules for model behavior, and whose values they encode; reasoning models

**Recommended Readings**

* OpenAI (2025), "Sycophancy in GPT-4o: What happened and what we're doing about it." https://openai.com/index/sycophancy-in-gpt-4o/
* OpenAI (2022), "Aligning language models to follow instructions." https://openai.com/index/instruction-following/
* Gibney (2025), "Secrets of DeepSeek AI model revealed in landmark paper," *Nature* (news). https://www.nature.com/articles/d41586-025-03015-6

**Session 10 – Mon, Oct 26 – Politics: Supply Side II — Public Institutions and Governance Capacity (Paci)**

**Class Topics**

- GenAI for public services: chatbots, triage, translation, drafting
- Administrative law and discretion; procurement and capacity
- Regulatory instruments: guidance, audits, liability, licensing, disclosure

**Recommended Readings**

* To be posted on Canvas

**Session 11 – Wed, Oct 28 – Tech: Computer Vision and Multimodality (Xu)**

**Class Topics**

- How a model reads a photo (patches) and a recording (spectrograms) as tokens; models that match photos with words
- What vision and speech models get wrong about people and politicians, and how researchers check them

**Recommended Readings**

* Trott (2024), "Vision-language models (VLMs), explained (pt. 1)," *The Counterfactual*, through the section on CLIP. https://seantrott.substack.com/p/vision-language-models-vlms-explained
* Jeon, Lee, Montgomery, and Lai (2026), "From Faces to Politics: Vision-Language Models (Sometimes) Link Visual Demographic Characteristics to Ideological Labels," *Political Analysis* 34: 613–621 (open access).

**Session 12 – Mon, Nov 2 – Politics: Data Centers (Paci)**

**Class Topics**

- Siting politics: local incentives, zoning, community opposition, water rights
- Energy and carbon footprint of training and inference; grid politics and disclosure
- Distributional effects and environmental justice

**Recommended Readings**

* To be posted on Canvas

**Session 13 – Wed, Nov 4 – Tech: Agentic AI: From Chatbots to Agents (Xu)**

**Class Topics**

- From tools to agents: models that plan, use tools, and act
- Delegation and oversight: who answers for what an agent does

**Recommended Readings**

* Schluntz and Zhang (2024), "Building effective agents," Anthropic (first half). https://www.anthropic.com/engineering/building-effective-agents
* METR (2025), summary post on measuring how long a task AI can complete. https://metr.org/blog/2025-03-19-measuring-ai-ability-to-complete-long-tasks/

**Session 14 – Mon, Nov 9 – Politics: US–China Competition (Paci)**

**Class Topics**

- Compute as a chokepoint: chips, supply chains, export controls
- Security and information operations; standards diplomacy
- Strategic dependence and technological alliances

**Recommended Readings**

* To be posted on Canvas

**Session 15 – Wed, Nov 11 – Tech: AI for Politics: Democratic and Authoritarian Uses (Xu)**

**Class Topics**

- AI in campaigns, policymaking, and civil society, in democracies and autocracies
- The 2026 midterms: what AI did, and what the evidence shows

**Recommended Readings**

* Kreps and Kriner (2023), "How AI Threatens Democracy," *Journal of Democracy* 34(4): 122–131.
* Yang and Roberts (2023), "The Authoritarian Data Problem," *Journal of Democracy* 34(4): 141–150.
  Before class: post on Canvas one AI-made political item you saw during the campaign.

**Session 16 – Mon, Nov 16 – Politics: AI Sovereignty (Paci)**

**Class Topics**

- Opportunities and constraints for the Global South
- Data governance, language access, and state capacity
- Policy diffusion across jurisdictions

**Recommended Readings**

* To be posted on Canvas

**Session 17 – Wed, Nov 18 – Tech: AI Agents: Reliability, Safety, and Governance (Xu)**

**Class Topics**

- How agents fail: mistakes, manipulation, and goals that go wrong
- Who sets the rules for AI agents; the tech track in review

**Recommended Readings**

* Narayanan and Kapoor (2025), "AI as Normal Technology," Part I. https://knightcolumbia.org/content/ai-as-normal-technology
* Kokotajlo et al. (2025), "AI 2027," the summary. https://ai-2027.com/

**Thanksgiving Recess – Nov 23 & 25 – No Class**

**Session 18 – Mon, Nov 30 – Final Presentations (Paci & Xu)**

**Class Topics**

- Student presentations in themed panels
- Panel synthesis: shared definitions and comparable outcomes

**Recommended Readings**

* See Canvas assignment

**Session 19 – Wed, Dec 2 – Final Presentations (Paci & Xu)**

**Class Topics**

- Student presentations in themed panels
- Course wrap-up: what we know, what we do not, and what to watch

**Recommended Readings**

* See Canvas assignment